



1. ngOnChanges()

```
export class ChildComponent implements OnChanges {  
  @Input() data!: string;  
  
  ngOnChanges(changes: SimpleChanges) {  
    console.log('ngOnChanges:', changes);  
  }  
}
```

Explanation :

1. Called before ngOnInit and whenever input properties change.
2. Runs every time a bound @Input() value changes.



2. ngOnInit()

```
export class AppComponent implements OnInit {  
  ngOnInit() {  
    console.log('ngOnInit: Component initialized');  
  }  
}
```

Explanation :

1. Called once after the first ngOnChanges().
2. Best place for initialization logic (fetch data, set defaults, etc.).



3. ngDoCheck()

```
export class AppComponent implements DoCheck {  
  ngDoCheck() {  
    console.log('ngDoCheck: Change detection running');  
  }  
}
```

Explanation :

- Called during every change detection cycle.
- Lets you customize change detection.
- Use carefully (can be performance heavy).



4. ngAfterContentInit()

```
export class AppComponent implements AfterContentInit {  
  ngAfterContentInit() {  
    console.log('ngAfterContentInit: Content projected');  
  }  
}
```

Explanation :

- Called once after Angular projects external content (<ng-content>) into the component. Called during every change detection cycle.



5. `ngAfterContentChecked()`

```
export class AppComponent implements AfterContentChecked {  
  ngAfterContentChecked() {  
    console.log('ngAfterContentChecked: Content checked');  
  }  
}
```

Explanation :

- Called after every check of projected content.
- Runs multiple times after each change detection.



6. ngAfterViewInit()

```
export class AppComponent implements AfterViewInit {  
  @ViewChild('myDiv') myDiv!: ElementRef;  
  
  ngAfterViewInit() {  
    console.log('ngAfterViewInit: View initialized');  
    console.log(this.myDiv.nativeElement.innerText);  
  }  
}
```

Explanation :

- Called once after component's view (and child views) are initialized.
- Good for accessing @ViewChild elements.



7. ngAfterViewChecked()

```
export class AppComponent implements AfterViewChecked {  
  ngAfterViewChecked() {  
    console.log('ngAfterViewChecked: View checked');  
  }  
}
```

Explanation :

- Called after every check of component's view (and child views).



8. ngOnDestroy()

```
export class AppComponent implements OnDestroy {
  intervalId: any;

  ngOnInit() {
    this.intervalId = setInterval(() => console.log('Running...'), 1000);
  }

  ngOnDestroy() {
    clearInterval(this.intervalId);
    console.log('ngOnDestroy: Component destroyed');
  }
}
```

Explanation :

- Called just before Angular destroys the component.
- Perfect for cleanup (unsubscribe, remove event listeners, clear intervals).



Execution Order

Explanation :

- ngOnChanges= when input property changes (runs before OnInit)
- ngOnInit= once, after first ngOnChanges
- ngDoCheck= every change detection cycle
- ngAfterContentInit= once, after content projection
- ngAfterContentChecked= after every content check
- ngAfterViewInit= once, after view init
- ngAfterViewChecked= after every view check
- ngOnDestroy= before component is destroyed